12.9 A Fair Decision from a Biased Coin

WORKSHEET

The question

STEP I 2019, Question 11

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

? **Ask yourself** In one round, what is the probability of a decision? Of no decision? What is the largest value $2pq$ can take?

STAGE 1 Decision on the nth round

Covers: Part (i), round n

? **Ask yourself** In one round, what is the probability that the coins agree? A decision on round n needs $n-1$ disagreements and then an agreement.

STAGE 2 Decision by round n

Covers: Part (i), by round n

Pause and try What is the probability of no decision in the first n rounds? How large can $pq = p(1-p)$ be?

STAGE 3 Three people

Covers: Part (ii)

Ask yourself Round one decides when all three coins agree. Otherwise, if the majority showed heads, the odd coin is turned to heads: what is now needed in round two? And with a tails majority?

Pause and try Write everything in terms of $s = pq$, using $p + q = 1$ and $p^3 + q^3 = (p+q)^3 - 3pq(p+q)$.

What's really going on

Von Neumann's trick

The procedure in the question is quick but biased. Von Neumann's procedure is fair: toss twice, decide on HT or TH, repeat on HH or TT. Each attempt succeeds with probability $2pq$, so it takes on average $\frac{1}{2pq}$ attempts, which is slow when the coin is very biased (Figure 12.9.1). The question's procedure decides with probability $p^2 + q^2 \geq \frac{1}{2}$ per round, but its answer favours the hotel when $p > \frac{1}{2}$.

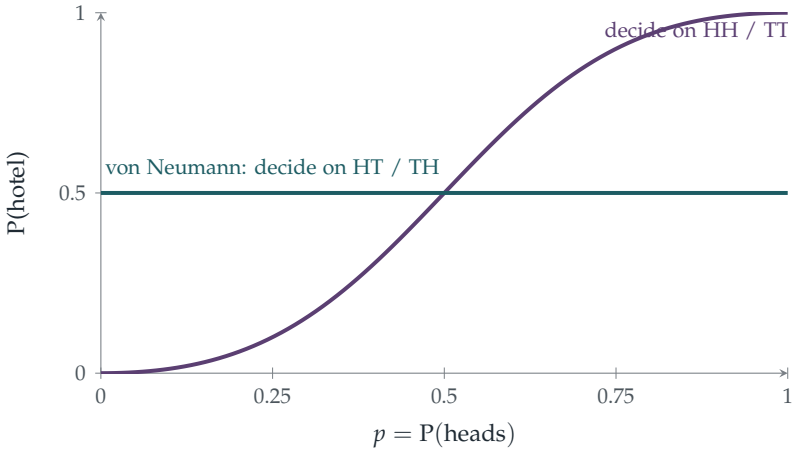


Figure 12.9.1 The probability of choosing the hotel, against the bias of the coin. Deciding when the coins agree (plum) is fast but favours the more likely face. Von Neumann's procedure (teal) is exactly fair for every p, at the cost of taking longer when the coin is very biased.

Extracting randomness

Von Neumann's trick is the first *randomness extractor*: a procedure that turns imperfect randomness into perfect randomness. It wastes a lot (each pair of tosses yields at most one fair bit, and often none), and better methods recycle the discarded HH and TT pairs to extract almost all the randomness the coin contains, which is measured by its *entropy* $-p\log_2 p - q\log_2 q$ bits per toss. Extractors are used in computers' hardware random number generators, which harvest unpredictable but biased physical noise and must turn it into unbiased bits for cryptography.

Von Neumann and random numbers

John von Neumann described the trick in a 1951 paper, "Various techniques used in connection with random digits", written when he and his colleagues were running some of the first computer simulations (the "Monte Carlo method") for the design of nuclear weapons. In the same paper he made a famous remark about generating random numbers by arithmetic: "Anyone who considers arithmetical methods of producing random digits is, of course, in a state of sin."

Try this next

1 **Expected rounds.** Find the expected number of rounds for the procedure in part (i), and for von Neumann's procedure. For which p is von Neumann's procedure slower?

2 **Fair three-way choice.** Using a biased coin, devise a procedure that chooses one of three options, each with probability exactly $\frac{1}{3}$.

3 **A biased die from a fair coin.** Using a fair coin, devise a procedure that simulates a coin with probability of heads exactly $\frac{1}{3}$. What is the expected number of tosses?

4 **Three people again.** In part (ii), what is the probability that they go to the hotel, as a function of p? Is the procedure fair when $p = \frac{1}{2}$?

Further reading

- **John von Neumann, "Various techniques used in connection with random digits"**, *National Bureau of Standards Applied Mathematics Series* 12 (1951). Short and famous.
- **Donald Knuth, *The Art of Computer Programming*, Vol. 2** (Addison-Wesley), Chapter 3, on random numbers.
- **Michael Mitzenmacher & Eli Upfal, *Probability and Computing*** (Cambridge University Press), on extracting randomness and entropy.

The full guided solution is in Section 12.9:
https://beyondstep.keithmcnulty.org/ch12/von-neumann.html